CS260 | Data Structures
Reading Assignment - Week 3

1 Instructions

- This assignment is worth 5 points.
- All the following readings are to be done from the book *Algorithms Illuminated by Tim Roughgarden*
- From *Chapter 12: Hash Tables and Bloom Filters*, read *section 12.1 (Supported Operations)*, *section 12.2 (Applications)*, and *section 12.3 (Implementation: High-Level Ideas)*
- **Take brief notes while reading**. Write your notes on a sheet of paper or on your computer.
- **Answer the questions below**. Write your answers on a sheet of paper or on your computer.
- **Submission:** Take a picture of your notes and answers (or a screenshot, or a pdf of the notes on computer), and upload it to gradescope.
- **Grading:** Grading will be based on effort, not correctness.

2 Questions

Question 1 : 2 points

Provide the notes you took for the readings.

Question 2 : 3 points

Answer the questions below:

(a) [.5 point] What do we mean when we say that a hash function is "good"?

(b) [.5 point] In the context of hashing, what are collisions?

(c) [.5 point] Collision Resolution, as the name suggests, is the idea of defining a procedure to resolve collisions in real-time. What are the two primary methods of collision resolution? Describe each method in a sentence.

(d) [1.5 points] Consider the following universe $\mathcal{U}$ of keys.

$$\mathcal{U} = \{xy \mid x \in \{a, b, c, d, e\} \text{ and } y \in \{1, 2, 3, 4, 5\}\}$$

(a) What is $|\mathcal{U}|$, the number of distinct keys in the universe $\mathcal{U}$?

(b) Let $|\mathcal{U}| = n$. Come up with a hash function that uses no more than $\sqrt{n}$ slots to store all keys in $\mathcal{U}$.